

September dataset and experiment update

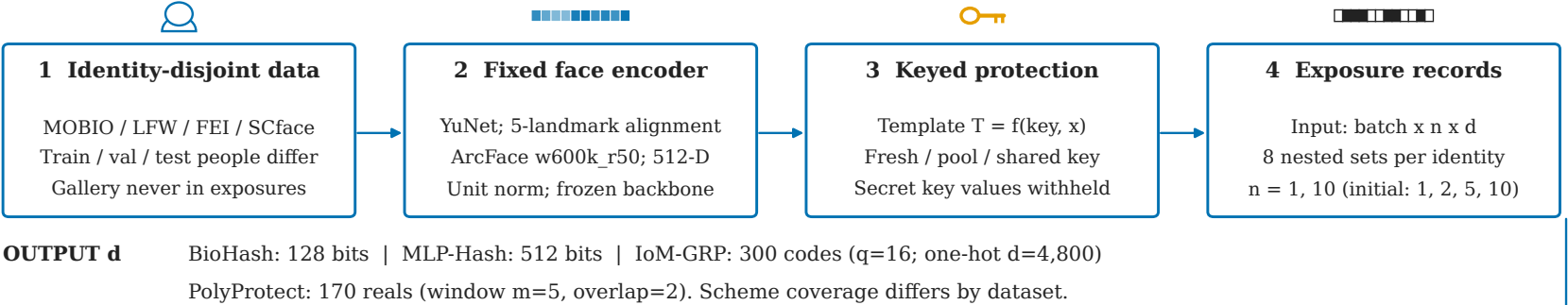
Identity-disjoint evaluation across session, pose and camera variation.

Dataset	Identities	Train / val / test	Valid / selected	Variation
MOBIO	150	90 / 30 / 30	1,799 / 1,800	Sessions
LFW	125	75 / 25 / 25	1,500 / 1,500	In-the-wild
FEI	200	120 / 40 / 40	2,378 / 2,400	Pose / expression
SCface	130	78 / 26 / 26	2,851 / 2,860	Camera / distance

SCface: mugshot-to-surveillance linkage under capture-domain shift.
Mugshot gallery / surveillance exposures; all 130 identities remain eligible after 9 detection failures.
Unprotected SCface top-1: 84.375% over 544 test probes; chance: $1/26 = 3.846\%$.
Evidence: 73 key-pool conditions / 12 studies; 72 one-seed model endpoints / 24 scheme pilot cells.
New MOBIO/FEI follow-up: 216 trained endpoints; 3 model seeds x 2 identity partitions.

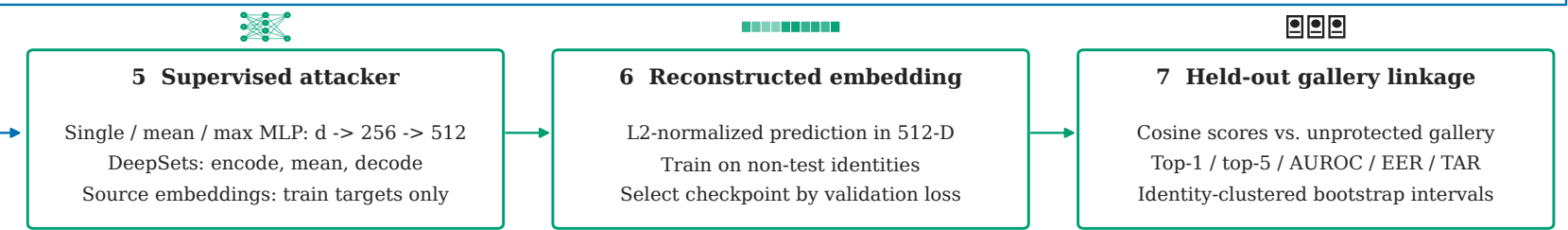
Experimental architecture

A DATA AND PROTECTION



B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

REPLICATION MOBIO / FEI follow-up: 2 identity partitions x 3 model seeds; matched 120-epoch training caps.

DESIGN Same-identity aggregation under controlled transform reuse; detailed attacker paths in the companion figure.

System overview: fixed encoder, hidden-key protection, same-identity exposure sets and held-out gallery linkage.

Detailed attacker: aggregation and reconstruction

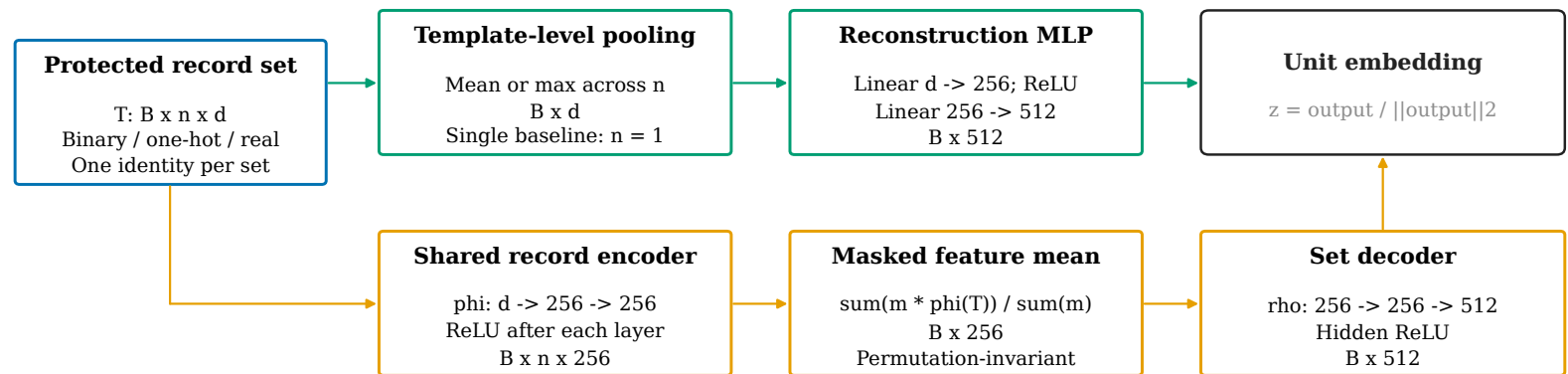
A MATCHED EXPOSURE CONSTRUCTION

One identity, n source records; nested subsets from one fixed permutation per repeat; gallery excluded.

Fresh: independent source-record keys. Pool K: hidden transforms recur across identity splits. No keys or slots in model input.

T: $B \times n \times d$ $d = 128$ BioHash; 512 MLP-Hash; 4,800 one-hot IoM-GRP; 170 PolyProtect

B TWO AGGREGATION PATHS



Primary: mean-pool MLP. Secondary: DeepSets. m masks valid records; all n records are valid in these runs.

C SUPERVISION

Target $u = \text{normalize}(\text{mean of exposed source embeddings})$.

Loss = $\text{mean}[1 - \text{cosine}(z, u)] + 0.1 \times \text{MSE}(z, u)$.

Adam: lr 0.001; weight decay 0.0001; hidden width 256.

Train identities only; select minimum validation loss; freeze test.

D HELD-OUT IDENTITY LINKAGE

Scores = $z \cdot G^T$; G: $N \times 512$ unprotected unit gallery.

Rank identities; top-1/top-5, AUROC, EER, TAR at FAR.

Follow-up: 3 model seeds x 2 identity partitions; 120 epochs.

Crossed seed/identity intervals; paired primary contrasts.

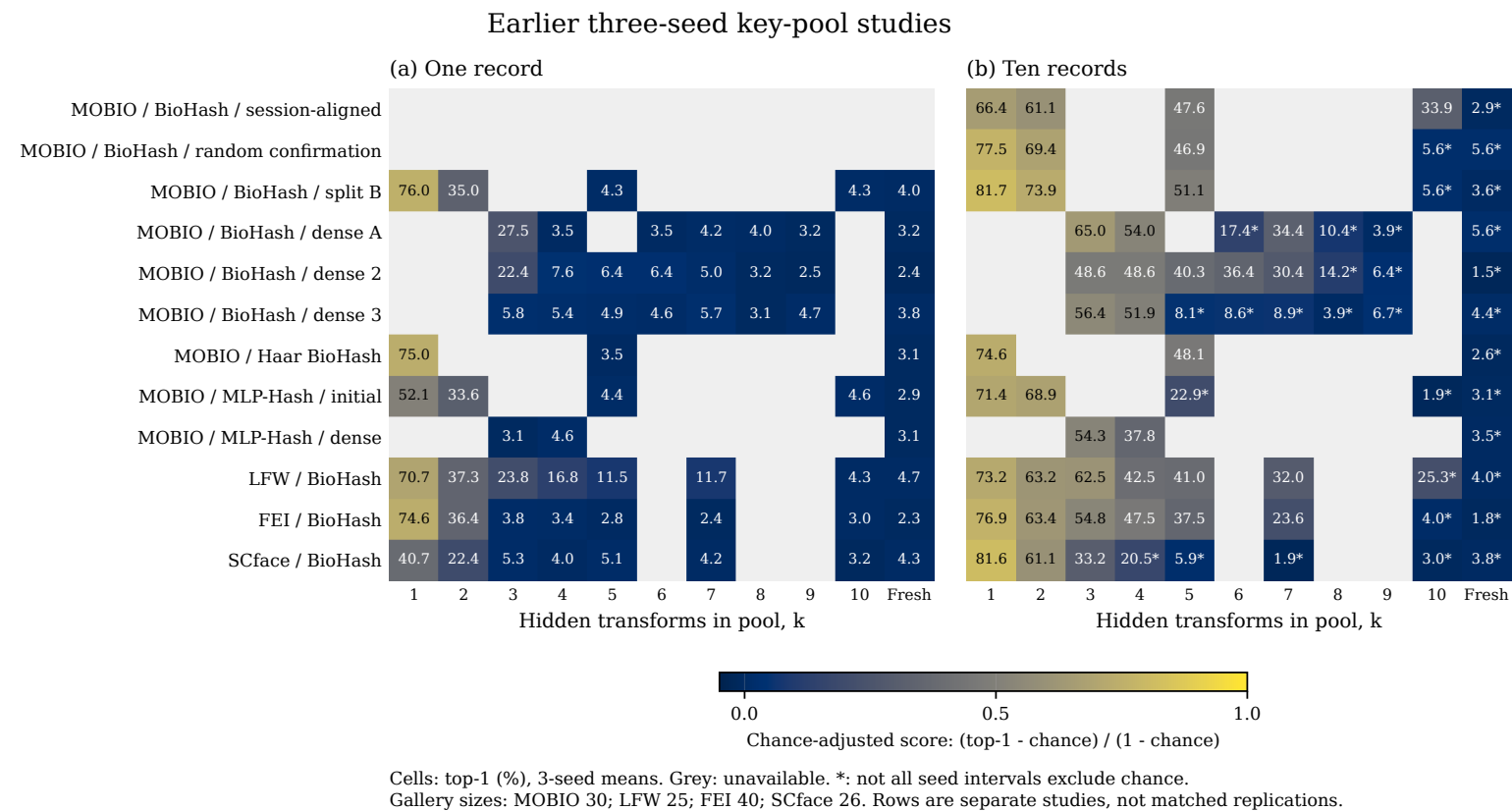
B: batch size; n: exposure count; d: template input width; N: test gallery identities. Source targets are not inference inputs.

Mean pooling is the primary endpoint; DeepSets is secondary. Target is the normalized mean of exposed embeddings, not the gallery.

Cross-dataset key-reuse boundary

SCface: pool 3 rises from 5.29% to 33.17% top-1; fresh keys give 3.85%.

SCface pools 1/2/3 pass the all-seed interval criterion; 4/5/7/10 fail. Failure is not equivalence.

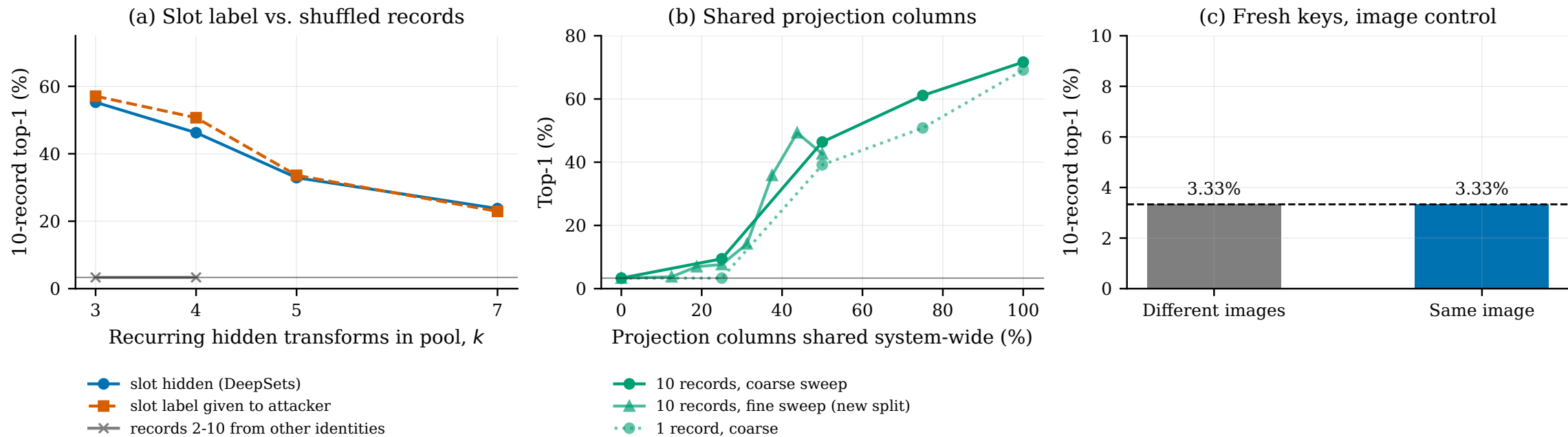


73 conditions / 12 source-separated, three-seed studies. Grey: unavailable. *: interval failure. Pilots excluded.

Mechanism controls and interpretation

MOBIO: slot labels change DeepSets by -0.83 to +4.44 points; shuffled records collapse to 3.33% chance.
Same-image fresh-key sets also give 3.33%. Projection-sharing effects depend on construction and partition.

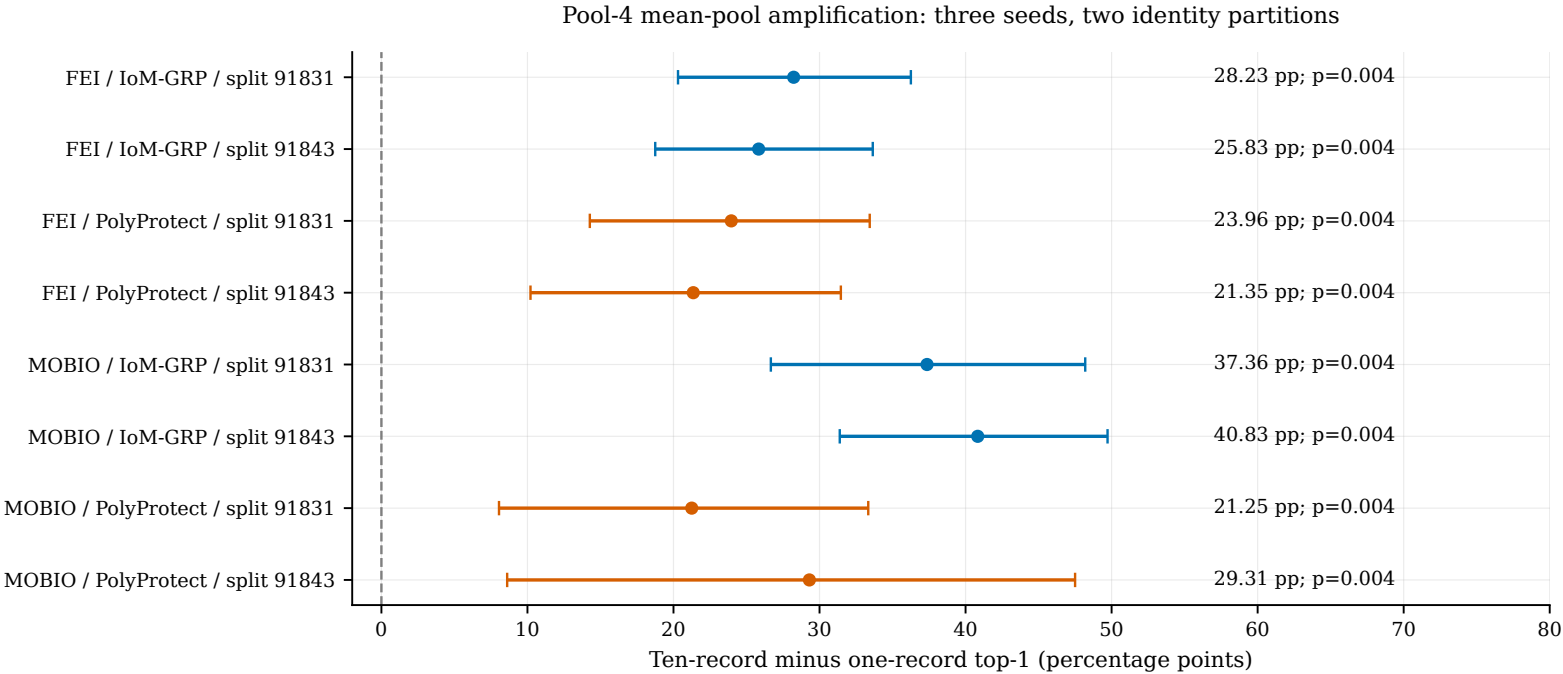
MOBIO mechanism controls (30 gallery identities; chance 3.33%)



Sources: experiments/mobio_mechanism_controls and mobio_correlation_controls. Three model seeds; chance 1/30.

Multi-seed, multi-partition validation

216 trained endpoints; fresh, shared and pool-4 keys; 3 model seeds x 2 identity partitions.
All 8 planned pool-4 amplification tests pass Holm correction (maximum adjusted $p = 0.004$).

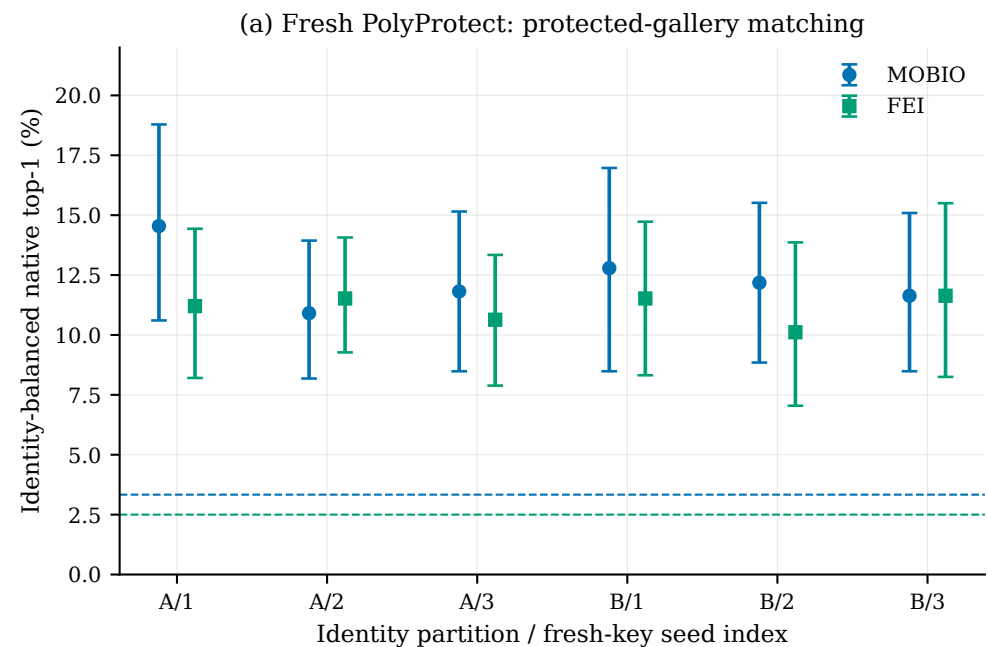


Bars: crossed model-seed / identity bootstrap 95% intervals. Tests: identity sign-flips, Holm family of 8.
Matched 120-epoch caps; key/set seeds fixed. Partitions are separate sensitivity analyses, not independent replications.

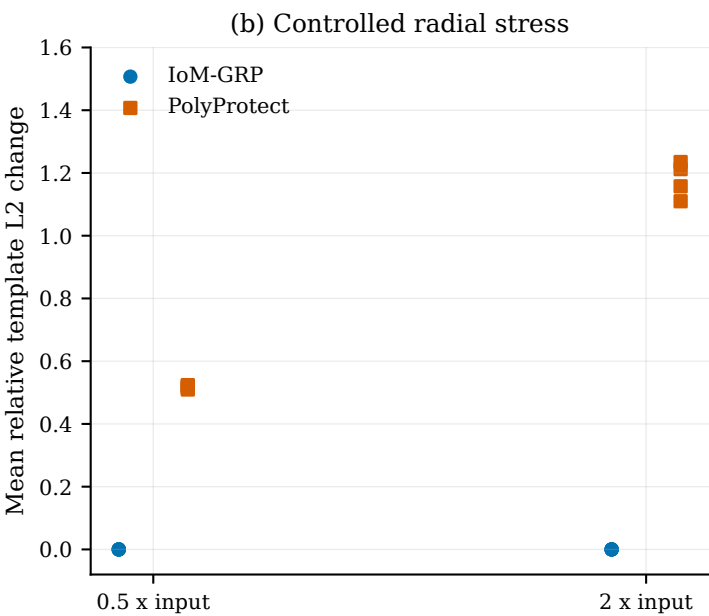
Matched 120-epoch caps. Crossed-bootstrap intervals include model-seed and identity resampling within each partition.

Native linkage and radial sensitivity

Fresh PolyProtect native identification exceeds the gallery-label null in 12 / 12 tests.
Maximum Holm-adjusted $p = 0.006$; three fresh-key seeds per dataset and identity partition.



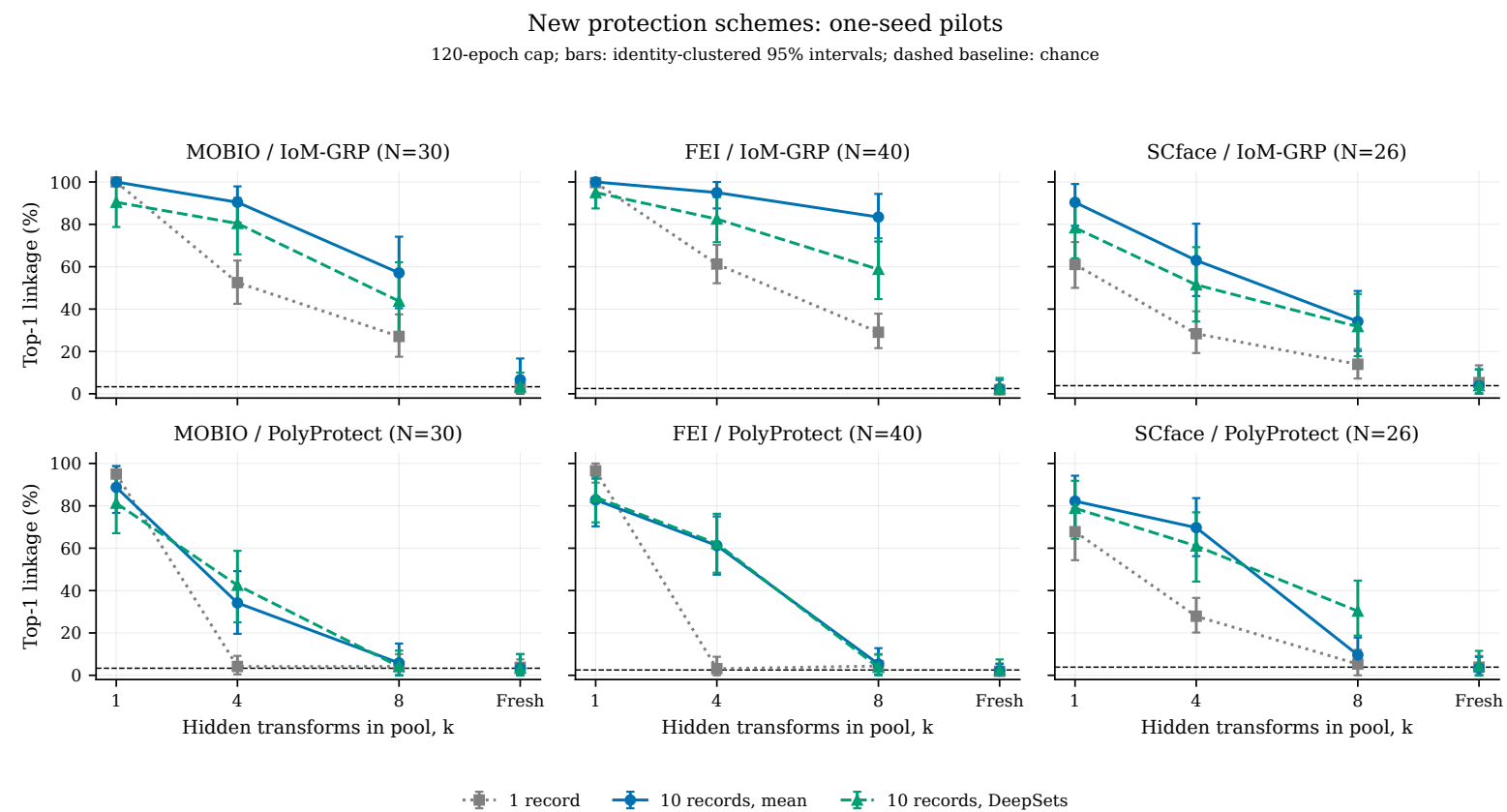
Native tests: 12 / 12 Holm-adjusted $p \leq 0.006$; identity-clustered 95% intervals; dashed lines: chance.
A/B: splits 91831/91843. Key seeds 1/2/3: 91873/91879/91883. Gallery-label permutations preserve probe clusters.
Radial stress: 32 held-out records per dataset/partition, fixed keys; measures scale sensitivity, not natural norm leakage.



Earlier synthetic stress: IoM-GRP unchanged, PolyProtect scale-sensitive. Genuine raw-input controls follow on page 12.

Earlier cross-dataset scheme pilots

Earlier one-seed pilots: 72 model endpoints on MOBIO / FEI / SCface, including pool 8.
Separate native fresh-key PolyProtect diagnostic: 12.73% / 13.73% / 10.66% top-1, respectively.



Pilot intervals condition on one seed and partition; not confirmation. New replicated MOBIO/FEI results are on page 6.

Closest research and our precise contribution

WHAT PRIOR WORK ALREADY SHOWED

Record-multiplicity attacks: several templates plus known parameters can enable recovery.

PolyProtect (2022): 1-10-record inversion; residual linkage under naive random parameters; stricter parameter selection improves unlinkability. Our native result is not a first discovery.

benchmark_cb: recognition, score-based unlinkability and information estimates across schemes.

FaceLinkGen v3: adaptive identity distillation from paired data with hidden per-query randomness.

WHAT OUR CONTROLLED EXPERIMENT ADDS

Different-image set aggregation with hidden fresh/shared/pool keys and disjoint training identities.

A measured one-to-ten benefit conditional on transform reuse, plus mechanism and failure controls.

Matched raw/shuffled/fixed-radius inputs separate scale effects from a norm-leakage claim.

NOT CLAIMED

Multiplicity, identity distillation and IoM scale invariance are not new.

No head-to-head accuracy win, exhaustive priority claim or break of stricter PolyProtect policies.

Sources: 10.1002/cpe.3042; 2110.00434v3; 2302.13286; 2602.02914v3. Full comparison: docs/literature/closest_work_2026-09-18.md.

Attacker access: assumptions and limits

SAME-PERSON RECORDS

An account or session pseudonym could group retained records without revealing real identity.
Stable grouping and retention are assumed; their prevalence in deployed products is not measured.

PAIRED TRAINING EXAMPLES

An authorized enrollment/query interface or a provider seeing inputs and outputs could supply pairs.
Crucially, training must use the SAME realized hidden pool as the target records.
A proxy with unrelated keys is insufficient. Identity splits are disjoint; recurring keys are not.

REFERENCE GALLERY

Lawful public or consented reference images; gallery images are excluded from exposure sets.
Only 25-40 candidate people, with guaranteed target membership: a small closed-set experiment.

BOUNDARIES

No open-set search, internet-scale distractors, unknown grouping or cross-provider transfer test.
Fresh independent transforms can also harm legitimate matching; they are not a tested defense.

Separate implementation and matching checks

FORMULA AND PARAMETER CHECKS

PolyProtect: separate scalar windows, zero padding, coefficients and powers match local outputs.

IoM-GRP: explicit grouped dot-product/argmax reference agrees; paper dimensions are tested.

Original PolyProtect uses different encoders and offers a stricter parameter-selection policy.

NATIVE MATCHING AUDIT: 48 MATCHED CELLS

Scalar PolyProtect and production outputs: zero difference after float32 casting.

Matrix and SciPy cosine scores: maximum difference $1.33\text{e-}15$; identical predictions.

Zero gallery/probe overlap, score ties, reversed-gallery changes or duplicate fresh keys.

The unit-input results reproduce the earlier three-key native averages.

RAW EXTRACTION CHECK

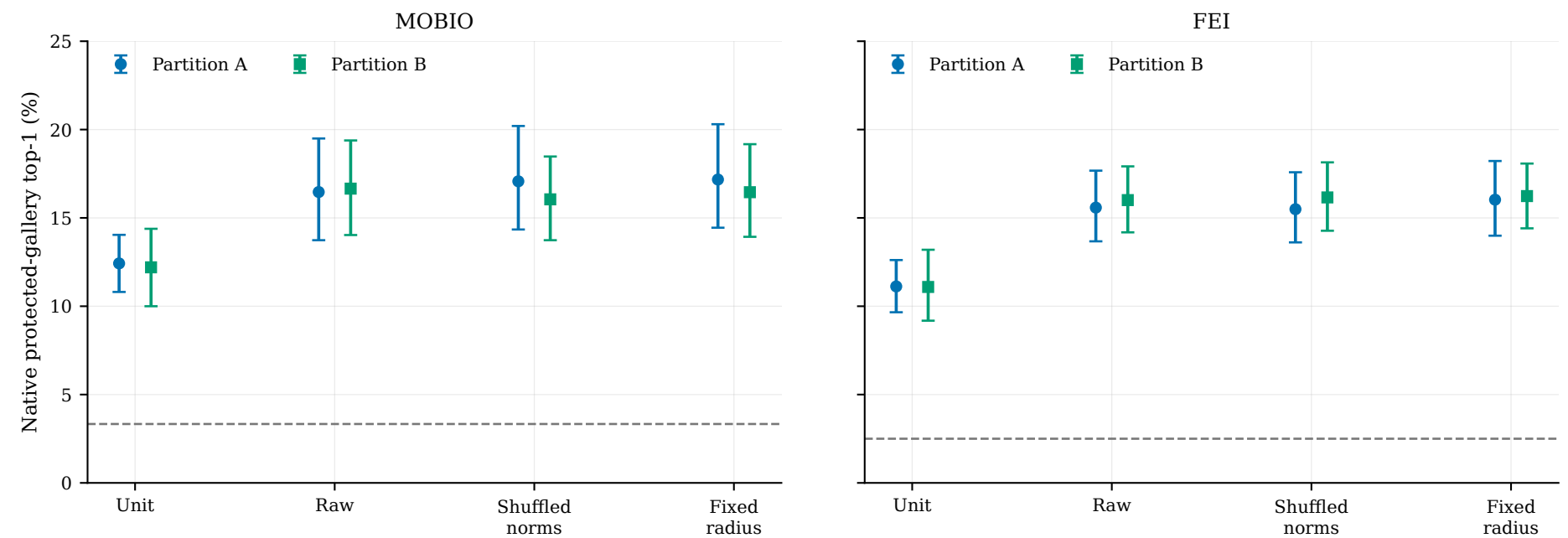
All 4,177 MOBIO/FEI records re-extracted; normalized discrepancy at most $2.98\text{e-}8$.

Natural raw/unit scaling changes zero of 38,400 sampled IoM codes.

Separate computational checks, not independent human review or official-code certification.

Raw embeddings: scale is not identity leakage

Real raw embeddings, shuffled norms and a fixed training-median radius; same keys in every arm.
Raw matching rises to 15.58-16.66%, but shuffled norms perform similarly; identity-specific norm leakage is not established.

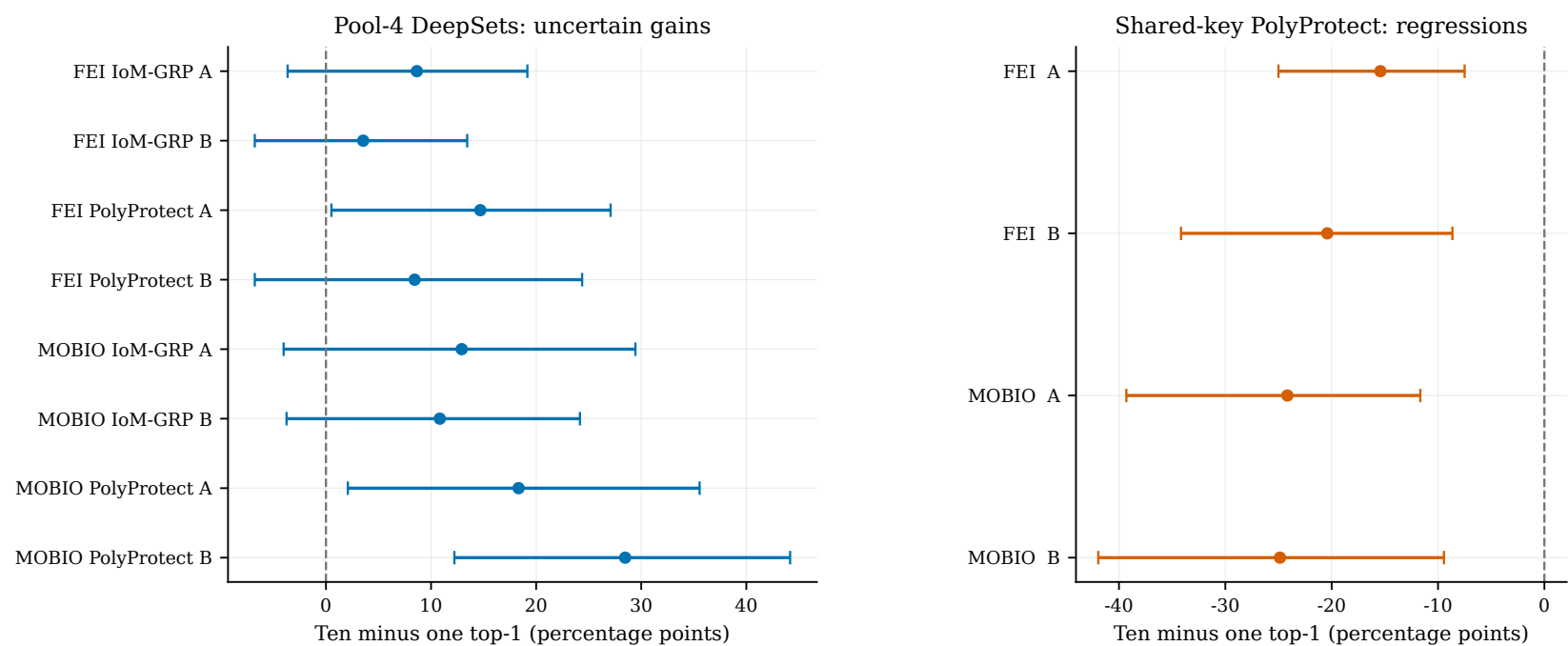


Three fixed key seeds averaged; identity-bootstrap 95% intervals. All 16 null tests: Holm $p = 0.0032$.
Raw minus unit survives paired correction only on FEI. Raw minus shuffled: all adjusted $p \geq 0.7584$.
Natural identity-specific norm leakage is not established. No learned raw-input attack was retrained.

Raw-unit gain survives correction only on FEI. Source-norm oracle: no corrected detection. No learned raw-input retraining.

Failure analysis: all 48 contrasts

Post-hoc two-sided family of 48; original primary eight-test family remains unchanged.
Eight pool-4 mean gains survive; four shared-key PolyProtect DeepSets losses also survive (adjusted $p = 0.0192$).



All 48 contrasts reported; post-hoc two-sided Holm family 48. Pointwise crossed seed/identity 95% intervals.
Left: none survives correction (minimum $p = 0.0576$). Right: all four negative contrasts survive ($p = 0.0192$).
All eight pool-4 mean-pool gains remain significant ($p = 0.0192$). More records do not universally improve linkage.

Seed ranges and leave-one-seed-out gains are published. Negative results constrain the claim: more records are not always better.

Findings and study scope

TRANSFORM REUSE AND RECORD MULTIPLICITY

Recurring hidden transforms support substantial identity linkage from multiple protected records.
The transition depends on dataset and identity partition, rather than a universal pool-size threshold.
SCface pool 3: 5.29% single-record versus 33.17% ten-record top-1; chance is 3.846%.

BOUNDARIES AND FAILURES

Shuffled-identity controls constrain the mechanism; same-pool training access remains essential.
Shared-key PolyProtect DeepSets loses 15.42-24.86 points: aggregation is not universally beneficial.
Native matching survives separate checks, but natural identity-specific norm leakage is not established.

STATISTICAL AND THREAT-MODEL SCOPE

BioHash evidence spans four datasets and three model seeds per study.
New scheme follow-up: 216 endpoints; all 8 primary contrasts pass Holm correction.
Native protected-gallery matching and learned embedding linkage measure different attack surfaces.
Fresh-key learned intervals include chance; they do not establish equivalence or general unlinkability.

Scope: fixed training key/set seeds, overlapping assignments, one encoder, small closed galleries, strong paired access.
Raw-input learned retraining, stricter PolyProtect policies and independent human review remain outside this revision.